NGUYEN VAN HUNG

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SUMMARY

Computer Science student seeking an AI Engineer Internship, with a strong foundation in Machine Learning, Deep Learning, and data analysis. Hands-on experience in model development, performance optimization, and deploying lightweight AI models on resource-constrained devices.

EDUCATION

University of Science (VNU-HN) **09/2024 – 8/2028 (Expected)**

Major: Computer Science

Relevant Coursework: Data Structures , Object-Oriented Programming, Machine Learning, Deep Learning

Achievements

- Ranked among the top 10 teams in the DATAFLOW Competition organized by the University of Science (VNU-HN), competing against 70 teams from universities across Northern Vietnam
- Earned an A+ in a specialized internship focused on machine learning and data processing

EXPERIENCE

Institute of Physics, Vietnam Academy of Science and Technology **Research Intern | 5/2026 – Present**

Contributed to the research and development of AI-integrated electronic devices by designing hardware and integrating sensors; collected, processed, and analyzed real-world data using digital signal processing techniques, improving analysis accuracy and developed and optimized deep learning models through compression for on-device inference.

PROJECT

Research on an Acoustic-Based Durian Ripeness Classification System **5/2026 - Present**

Video demo:https://www.youtube.com/watch?v=DyJ836IZEWA

Repo: https://github.com/nghungvkck/Durian-In-esp32-Edge-AI-

- Data Collection & Signal Processing: Collected 1,126 acoustic samples from 28 durians using ESP32-S3 and ICS-43434 microphone. Developed an audio processing pipeline with signal smoothing, peak detection, event segmentation, and log-Mel Spectrogram extraction (128 × 16). Applied data augmentation techniques to address class imbalance.
- Model Development & Optimization: Achieved 94.90% test accuracy (F1-score: 0.95). Applied Knowledge Distillation and structured pruning, reducing model parameters by 98.2% (8.91M → 0.158M) while maintaining accuracy. INT8 quantization achieved 86.67% accuracy.
- Edge Deployment: Developed firmware using ESP-IDF and deployed an 181.83 KB ESP-DL model on ESP32-S3, achieving an average inference latency of 343.56 ms per audio segment.

Autoscaling Forecasting & Optimization **1/2026 - 4/2026**

Video demo: https://www.youtube.com/watch?v=bJVfUWz4bss

Repo: https://github.com/nghungvkck/AUTOSCALING_ANALYSIS

- Data Engineering & Feature Engineering: Processed 2.9M Apache HTTP access-log requests, performing data cleaning, aggregation, and time-series transformation. Engineered 13 features covering temporal patterns, lag dependencies, burst detection, and rolling statistics across multiple time windows.
- Forecasting & Autoscaling Optimization: Developed XGBoost, LightGBM, and Hybrid LSTM-Prophet models, achieving $R^2 = 0.94$ and MAPE = 7.2%. Designed a hybrid autoscaling strategy with estimated 25–35% cost reduction and over 99.2% SLA compliance.
- System Development & Simulation: Built a Streamlit dashboard and FastAPI services for forecasting visualization, autoscaling simulation, and cost comparison across Hybrid, Reactive, and Fixed strategies.

SKILLS

- Programming & Data: Python, C/C++, SQL, NumPy, Pandas, Scikit-learn
- AI/ML & Backend: PyTorch, FastAPI, Flask, NoSQL
- Infrastructure & Tools: Docker, Git/GitHub, Linux, ESP-IDF